TMW-27 Floorstander

Follow this guide to build the cabinet: tweeter, 4-inch midrange and 8-inch woofer, with a separate sealed chamber for the midrange and a 46-litre vented chamber for the woofer. All dimensions are in millimetres. The drawings are to scale but not to a fixed print scale, so always work from the written numbers.

SIZE W×D×H	PANELS	WOOFER CHAMBER	MID CHAMBER
270×340×1040	18 mm birch ply	45.9 L net	15.7 L sealed
PORT	**PORT TUNING**	**BASS -3 DB**	**CROSSOVERS**
Ø76 × 349	27 Hz	31 Hz*	350 / 2600 Hz

* With the 20 Hz protective filter, measured in free space. The box alone reaches 27 Hz, and a room usually adds bass below 60 Hz.

What is inside

ZONE	FROM TOP (MM)	CONTENTS
Top panel	0 – 18	Solid
Mid chamber	18 – 252	Tweeter, midrange, filled with wool
Divider	252 – 270	Airtight panel
Woofer chamber	270 – 1022	Woofer, 2 braces, port, wall lining
Bottom panel	1022 – 1040	Port exits here
Gap + plinth	1040 – 1105	40 mm gap, 25 mm plinth

Before you cut

- Buy the drivers first. Check their real cut-out diameter, flange size and depth against pages 2 and 3.
- Enter the woofer's real datasheet values in the box simulator (web/index.html). If it gives a different volume or port length, change the woofer chamber height or the port length to match.
- Build the second speaker as a mirror image: the tweeter offset goes toward the inner edge on both.
- Cut a test piece to check your saw and router settings before cutting the baffles.
- Wear eye and ear protection and use a dust mask. MDF and ply dust is harmful.

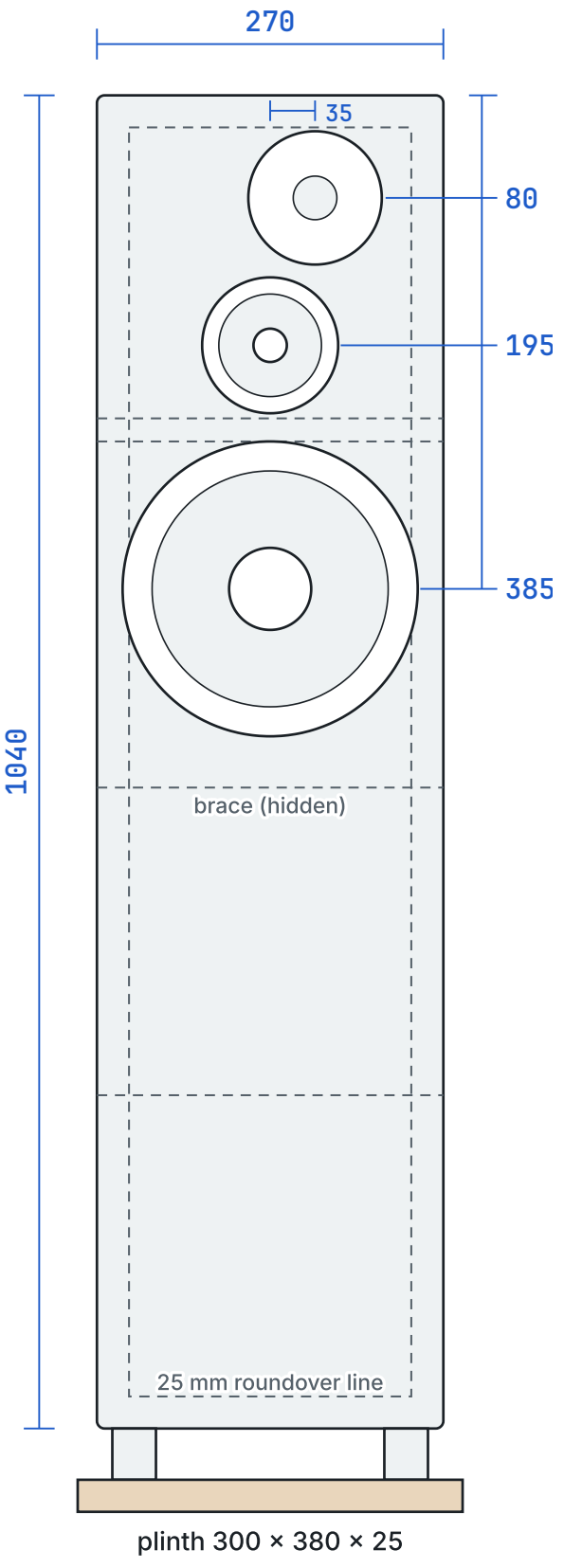
Airtight matters. A leak in the woofer chamber or the divider ruins the bass tuning. Seal every inside joint and every wire hole.

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DRAWING A

Front view and driver positions



DRIVER	CENTRE FROM TOP	ACROSS
Tweeter	80	170 from left*
Midrange	195	135 (centre)
Woofer	385	135 (centre)

* Left speaker as seen from the front. On the mirrored speaker the tweeter is 100 from the left (35 mm off-centre toward the inner edge).

Write in your driver sizes

DRIVER	CUT-OUT Ø	FLANGE Ø	FLANGE DEPTH
Tweeter			
Midrange			
Woofer			

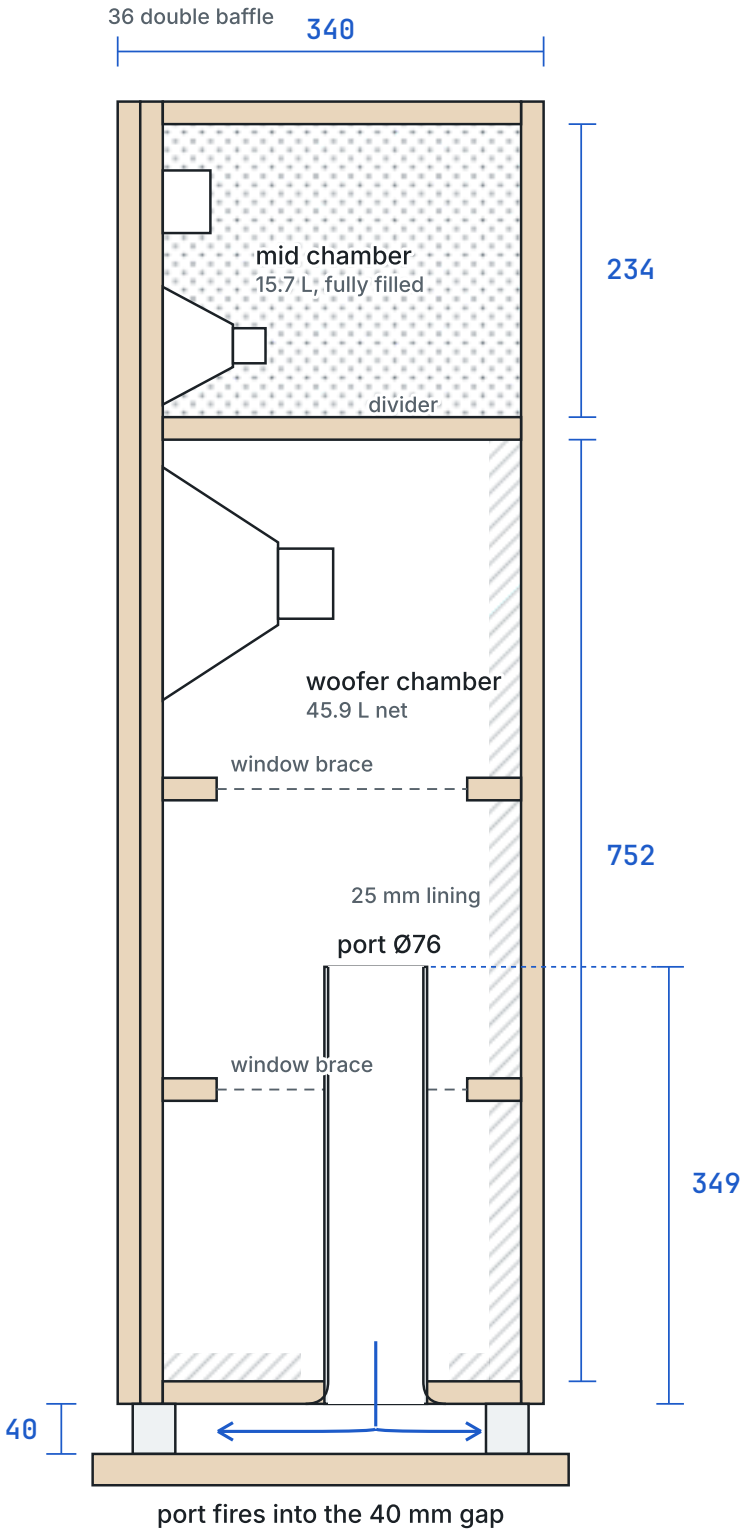
Baffle routing

- **Outer layer:** cut-out hole, plus a recess the size of the flange Ø and as deep as the flange, so each driver sits flush.
- **Inner layer:** cut-out hole only, then chamfer its back edge at 45° for the woofer and midrange so air can move freely behind the cone.
- The tweeter only needs its cut-out through both layers.
- After gluing the two layers, rout a **25 mm roundover** on the two vertical front edges and the top front edge.
- Mount the drivers with M5 T-nuts and bolts, not wood screws.

> Driver spacing check: tweeter bottom edge to mid top edge ≈ 10 mm, mid bottom edge to woofer top edge ≈ 22 mm (with a 104 mm tweeter, 106 mm mid and 230 mm woofer flange). If your flanges are bigger, move the woofer down, not the tweeter.

DRAWING B

Side section, chambers and port



PART	HEIGHT FROM TOP
Top panel	0 – 18
Divider	252 – 270
Upper brace	540 – 558
Lower brace	780 – 798
Bottom panel	1022 – 1040
Port top end	691

Inside dimensions

Width	234
Depth	286
Mid chamber height	234
Woofer chamber height	752

Port

- Ø76 mm inside, flared tube. Total length 349 mm, measured from the underside of the bottom panel.
- Cut it 360 mm long and trim to tune (page 8).
- Keep at least 76 mm of open space around its top end.
- Round the hole on the underside of the bottom panel with about an 18 mm radius.

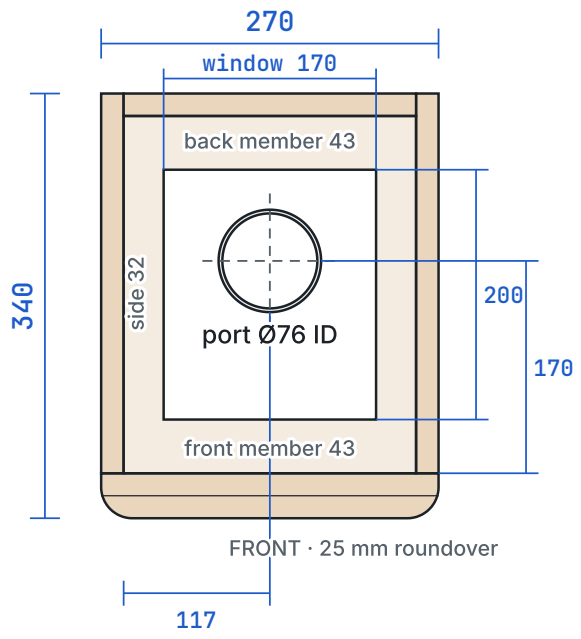
Damping

- **Woofer chamber:** butyl or bitumen pads on the sides and back, then 25 mm felt on the sides, back and floor. Leave the middle open.
- **Mid chamber:** fill it loosely with wool or polyester, about 3/4 full.

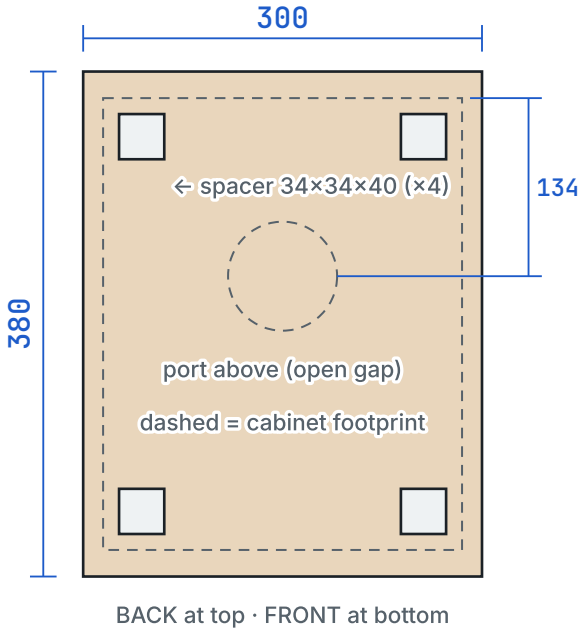
DRAWING C

Brace, port position and plinth

C1 · PLAN SECTION THROUGH A WINDOW BRACE (LOOKING DOWN)



C2 · PLINTH, TOP VIEW, SPACER POSITIONS



BACK at top · FRONT at bottom

Window braces (×2)

- Outer size 234 × 286, the same as the inside of the box.
- Window 170 wide × 200 deep, leaving 32 mm side members and 43 mm front and back members.
- Glue all four edges to the baffle, sides and back. The brace only stiffens the walls if it is glued on every side.
- The port tube passes through the lower brace window. Its centre is 135 mm from either outer side and 170 mm behind the inner face of the baffle (206 mm from the front of the speaker).

Plinth and gap

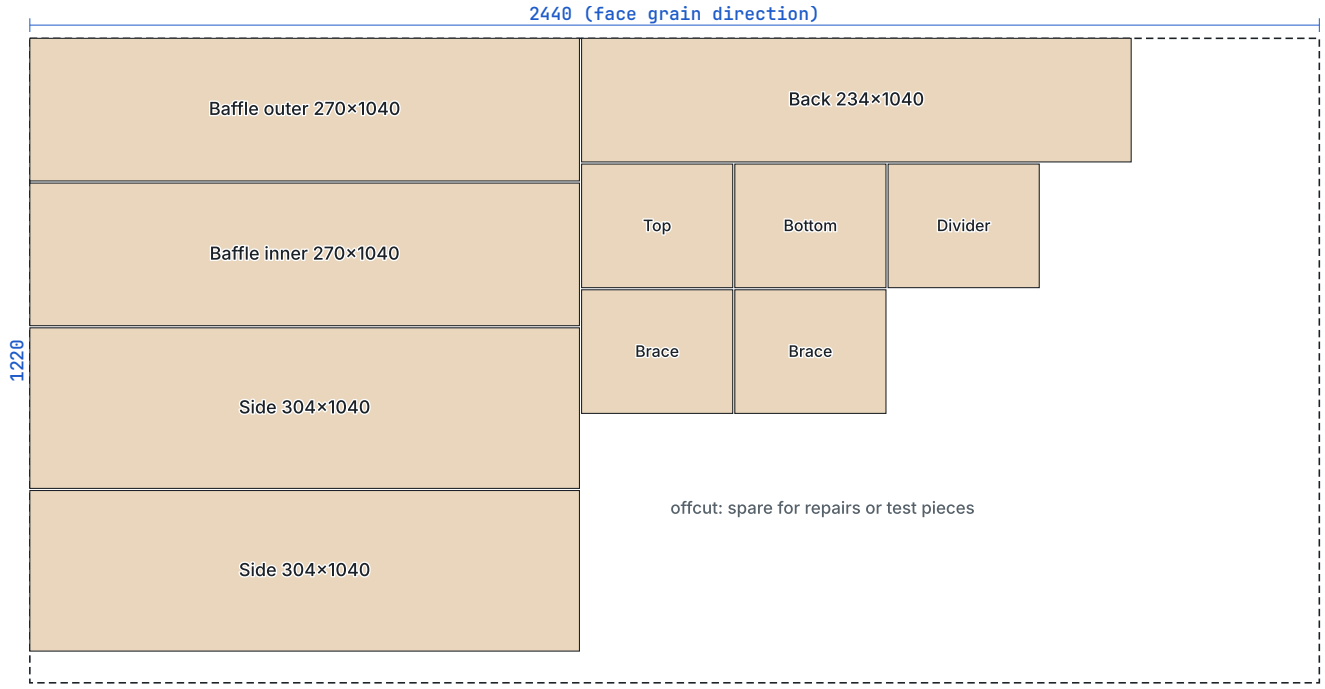
- Plinth 300 × 380 × 25 mm, overhanging the cabinet by 15 mm each side and 20 mm front and back.
- Four hardwood spacers, 34 × 34 × 40 mm, inset 12 mm from the cabinet corners. Glue or screw them to the cabinet and screw the plinth on.
- The 40 mm gap must stay open on all four sides, because the port breathes through it. Never stand the speaker on carpet without the plinth.
- Fit spikes or rubber feet under the plinth.

STAGE 1

Cut list and sheet layout (one speaker)

#	PART	QTY	SIZE (MM)	NOTES
1	Baffle, outer layer	1	270 × 1040 × 18	Holes and flange recesses, roundover after gluing
2	Baffle, inner layer	1	270 × 1040 × 18	Holes, chamfered on the back
3	Sides	2	304 × 1040 × 18	Glued behind the baffle
4	Back	1	234 × 1040 × 18	Between the sides. Terminal holes in the woofer chamber
5	Top, bottom, divider	3	234 × 286 × 18	The bottom has a Ø82 hole for the port (check your tube's outer Ø)
6	Window braces	2	234 × 286 × 18	170 × 200 window
7	Plinth	1	300 × 380 × 25	From a separate board
8	Spacers	4	34 × 34 × 40	Hardwood
9	Port tube	1	Ø76 ID × 360	Trim to 349

ONE 1220 × 2440 × 18 MM SHEET PER SPEAKER · 4 MM SAW KERF ALLOWED BETWEEN PARTS



> Cut the long parts along the 2440 direction so the face grain runs vertically on the finished speaker. Check that the sheet is square first, then cut every panel of the same size at one fence setting so all matching panels come out identical.

STAGE 2

Parts and materials (for the pair)

ITEM	QTY (PAIR)	RECOMMENDED	ALTERNATIVES / NOTES
Plywood 18 mm	2 sheets	Baltic birch, void-free, 1220×2440	MDF 18–19 mm on a budget. Seal the edges in humid rooms
Board 25 mm	2 pcs	300×380 ply or MDF	For the plinths
Hardwood block	8 pcs	34×34×40	Spacers
Woofer 8"	2	Dayton Audio RS225-8	SB Acoustics SB23 series, Scan-Speak 22W. Re-check the box size
Midrange 4"	2	Dayton Audio RS100-8	SB Acoustics SB12 midrange
Tweeter 1"	2	SB Acoustics SB26STAC-C000-4	Dayton Audio RST28F-4. Fs under 800 Hz
Port tube	2	Ø76 mm (3") flared, ≥ 360 long	PVC pipe with routed flares
DSP + amplifier	2 or 1	Hypex FusionAmp FA253, one per speaker	Dayton DSP-408 plus a 6-channel class-D amp, or ADAU1701 4-ch DSP amp boards
Measurement mic	1	miniDSP UMIK-1 with REW (free)	Dayton DATS V3 for impedance (optional)
Terminals	6 pairs	Binding posts, or 1 Speakon each	3 pairs per speaker for active drive
Wire	~6 m	1.5 mm² (16 AWG) copper	Label each pair
Damping pads	~1.2 m²	Butyl or bitumen, 2–3 mm, self-adhesive	On the woofer chamber sides and back
Felt lining	~1.6 m²	25 mm polyester or wool felt	Woofer chamber walls
Fill	~0.5 kg	Sheep's wool or polyester fibre	Mid chambers
Fixings	~40	M5 T-nuts and hex bolts	Count your driver holes
Glue and sealant	—	PVA (Titebond II/III), acrylic sealant, hot glue	Gasket tape under every driver
Tools	—	Table or track saw, router + circle jig, 25 mm and 18 mm roundover bits, clamps	Many timber shops will cut panels to size for you

STAGE 3

Build steps

- [ ] 01 Cut all panels (page 5). Label each one and mark which speaker it belongs to, left or right.
- [ ] 02 Clamp the two baffle layers together. Drill pilot holes at the driver centres: 80, 195 and 385 mm from the top, with the tweeter 35 mm off-centre (mirrored on the second speaker).
- [ ] 03 Rout the cut-outs through both layers with the circle jig. Rout the flange recesses in the outer layer only.
- [ ] 04 Chamfer the back of the woofer and mid holes 45° in the inner layer. Fit the T-nuts from the back.
- [ ] 05 Glue the two baffle layers together (spread glue thinly and evenly). Clamp flat and leave overnight.
- [ ] 06 Cut the windows in the two braces and the port hole in the bottom panel. Round the underside of the port hole.
- [ ] 07 Glue one side to the baffle, square it up, then glue in the top, divider, braces and bottom at the heights on page 3. Then the second side.
- [ ] 08 Glue the port tube through the bottom panel and the lower brace window. Its top end should be 349 mm above the underside of the bottom panel, or longer for now.
- [ ] 09 Drill the terminal holes in the back panel (woofer chamber, about 150 mm above the bottom) and a wire hole in the divider.
- [ ] 10 Glue on the back. Once dry, run sealant along every inside joint you can reach through the driver holes, especially around the divider.
- [ ] 11 Rout the 25 mm roundover on the front edges. Sand, then finish (veneer, oil or paint). Mask the driver recesses.
- [ ] 12 Fit the damping pads and the 25 mm felt in the woofer chamber. Keep the area above the port clear.
- [ ] 13 Run the wires: woofer, mid and tweeter pairs to the terminals. Seal the divider hole and the terminal holes with hot glue.
- [ ] 14 Fill the mid chamber with wool. Put gasket tape on each driver flange and mount the drivers with bolts. Tighten evenly in a star pattern, finger-tight plus a quarter turn.
- [ ] 15 Leak test: play a 20–30 Hz tone at low level and listen or feel around the joints, terminals and drivers. Air should only come out of the port.
- [ ] 16 Fix the spacers and the plinth. Check that the 40 mm gap is open all round.

STAGE 4

Tuning and setup

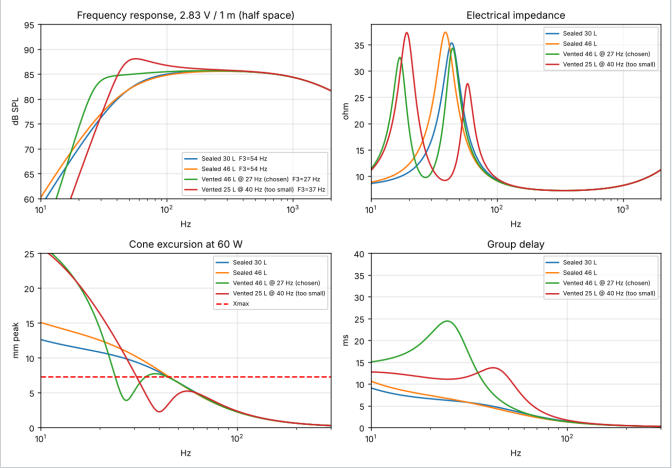
DSP CHANNEL	FILTERS	START GAIN
Woofer	High-pass 20 Hz LR 24 dB/oct (protection) · low-pass 350 Hz LR 24 dB/oct · low shelf +3 to +6 dB at ~425 Hz (baffle step)	0 dB
Midrange	High-pass 350 Hz LR 24 dB/oct · low-pass 2600 Hz LR 24 dB/oct · same low shelf	0 dB
Tweeter	High-pass 2600 Hz LR 24 dB/oct	-5 dB
All	All drivers in the same polarity. Measure the delays (tweeter usually 0.1–0.3 ms)	–

- [ ] T1 **Check the port tuning before any other step.** Measure the woofer's impedance. The dip between the two bass peaks should be at **27 Hz**. If it is higher, the port is too short, so fit a longer tube. If it is lower, trim 5 mm at a time. Write the result here: _____ Hz
- [ ] T2 Check the polarity of each driver with a 1.5 V battery: the cone moves forward when + is on the red terminal.
- [ ] T3 Put the mic 1 m away at tweeter height. Measure each driver alone and set the gains so all three bands are level.
- [ ] T4 Use REW's impulse alignment to set the mid and tweeter delays. Check for a smooth sum through 350 Hz and 2600 Hz, with no dip.
- [ ] T5 Set the baffle-step shelf: +6 dB far from walls, +3 to +4 dB if the speakers stand 30–60 cm from the back wall.
- [ ] T6 At your listening seat, only cut room peaks below 300 Hz. Don't boost dips, and leave the treble alone.
- [ ] T7 Place the speakers 50–80 cm from the back wall in an equilateral triangle with your seat, toed in toward you, tweeters near ear height (≈ 1025 mm with the plinth).

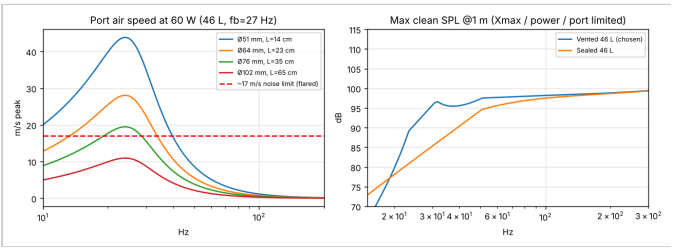
> Never run the woofer without the 20 Hz protection filter. Below the port tuning, the cone is no longer held back by the air in the box, and the simulation shows it passing its 7.3 mm limit at 60 W.

BACKGROUND

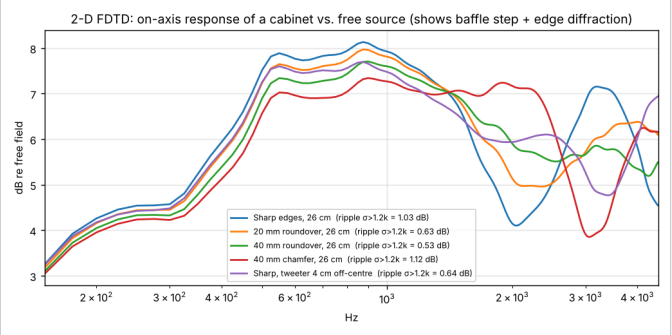
Why it is designed this way



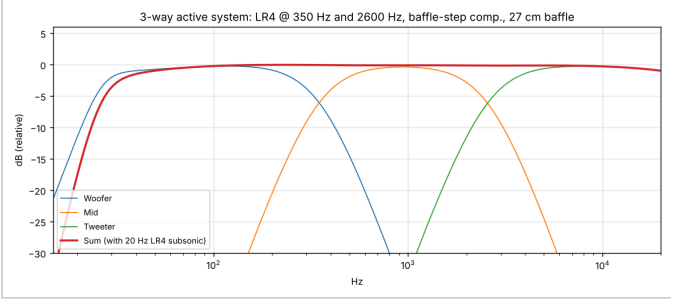
Sealed boxes stop at 54 Hz. The vented 46 L box reaches 27 Hz without a hump.



Smaller ports blow air too fast and chuff. Ø76 mm is the largest that still fits in the box.



Wave simulation: rounded edges and an off-centre tweeter halve the edge-diffraction ripple.



The full active 3-way system after DSP correction, flat within about ±0.6 dB.

Based on Thiele (1961) and Small (1973) for vented boxes, Olson (1951) for cabinet edge diffraction, Harwood and Iverson for panel resonance, and Salvatti/Devantier/Button and Bezzola for port design. The simulation code is in the repository under simulator/. Driver values are nominal, so measure your finished speakers.